

UME753 : Advanced Mechanical Vibrations

L	T	P	Cr
3	0	0	3.0

Course Objective: Introduction to Advanced Mechanical Vibrations starts with a review of single and double degree of freedom systems. After that, multiple degrees of freedom systems are introduced to explain the vibrations of string and beam. These vibration systems provide to apply or use them into practical problems.

Syllabus

Introduction: Review of free and forced vibrations with and without damping.

Isolation: Vibration isolation and transmissibility; Un-damped vibration absorbers.

Multi degree of freedom system: Generalized coordinates and coordinate coupling; Orthogonality of modes, Free and forced vibration of multi-degree of freedom systems with and without viscous damping; Lagrange's equation; Holzer's method. Solution of Eigen value problem, transfer matrix and modal analysis.

Stability criterion: Self excited vibrations; Criterion of stability; Effect of friction on stability. **Nonlinear vibration:** Free vibrations with non-linear spring force or nonlinear damping; Phase plane; Energy curves; Lienard's graphical construction; Method of isoclines.

Vibration of continuous system: Vibrations of strings; Free and forced longitudinal vibrations of prismatic bars; Ritz and Galerkin methods.

Random vibration: Mathematical descriptions of stochastic processes; Stationary and ergodicity; Gaussian random process, correlation functions and power spectral density.

Course Learning Objectives (CLO)

The students will be able to:

1. Ability to define, formulate and solve advanced problems of mechanical vibrations
2. Being able to provide information about determining Natural Frequencies and Mode shapes
3. Being able to provide information about determining forced response

Text Books

1. Rao, S.S., "Mechanical Vibrations", 4th Ed., Pearson Education., 2007
2. Meirovitch, L., "Fundamental of Vibrations", Mc-Graw Hill., 2001
3. Inman, D.J., "Vibration and Control", John Willey & Sons., 2002
4. Tamadonni, S. and Kelly, G.S., "Mechanical Vibrations", Mc-Graw Hill., 1998
5. Rao, J. S., "Vibration Condition Monitoring of Machines", Tata Mc-Graw Hill., 2006

Evaluation Scheme

S.No.	Evaluation Elements	Weightage (%)
	MST	30
	EST	45
	Sessional (May include Assignments/Quizzes/Tutorial Evaluations)	25